

NAME:AYUSH PATHAK CLASS:CSAID-B3 ROLL:60

```
import pandas as pd
import numpy as np

from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
import matplotlib.pyplot as plt
import seaborn as sns
```

▼ =====

2. LOAD DATASET

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```
df = pd.read_csv('/content/Iris.csv')

print(df.head())
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

▼ =====

3. DATA PREPROCESSING

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```
# Drop unnecessary column
df.drop(['Id'], axis=1, inplace=True)

# Features only (exclude species)
X = df.drop(['Species'], axis=1)

# Scaling (important for KMeans)
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

▼ =====

4. ELBOW METHOD

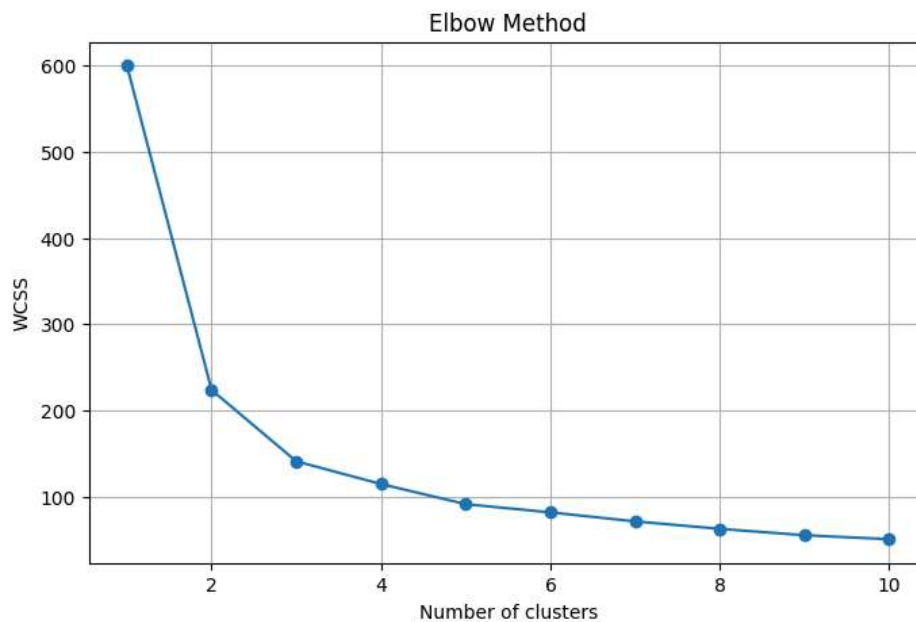
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```
wcss = [] # Within-cluster sum of squares

for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, init='k-means++', random_state=42, n_init=10)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

# Plot elbow graph
plt.figure(figsize=(8,5))
plt.plot(range(1, 11), wcss, marker='o')
plt.title('Elbow Method')
```

```
plt.xlabel('Number of clusters')
plt.ylabel('WCSS')
plt.grid()
plt.show()
```



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5. APPLY K-MEANS

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```
# From elbow -> optimal k = 3 (you'll see bend at 3)
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
y_kmeans = kmeans.fit_predict(X_scaled)

# Add cluster labels to dataframe
df['Cluster'] = y_kmeans

print(df.head())
```

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species \
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1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

	Cluster
0	1
1	1
2	1
3	1
4	1

=====

6. VISUALIZATION

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```
# Use 2 features for visualization
plt.figure(figsize=(8,6))
```

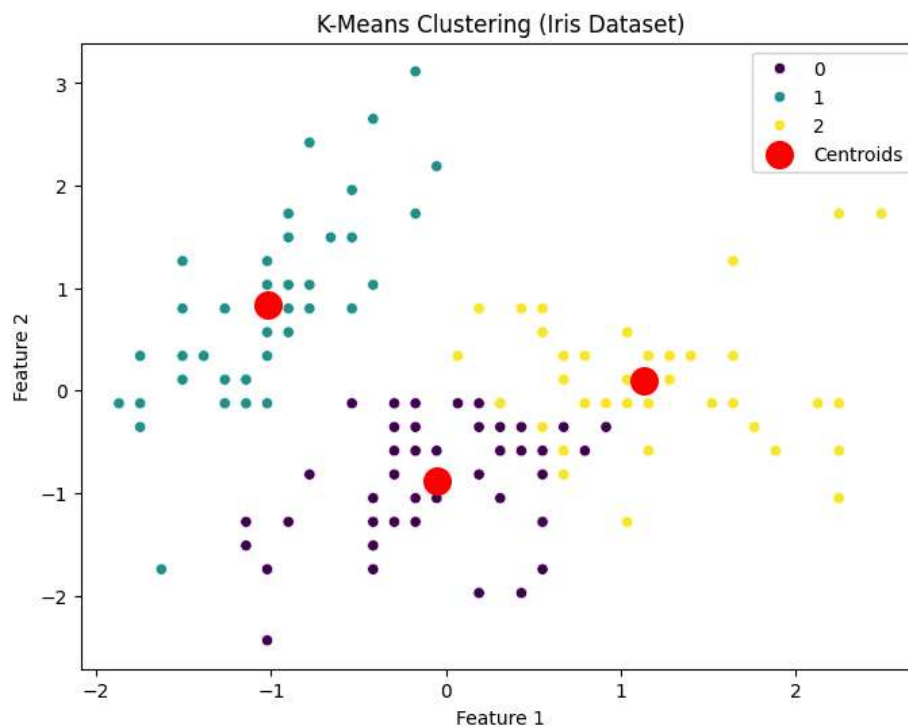
```

sns.scatterplot(
    x=X_scaled[:, 0],
    y=X_scaled[:, 1],
    hue=y_kmeans,
    palette='viridis'
)

# Plot centroids
centroids = kmeans.cluster_centers_
plt.scatter(
    centroids[:, 0],
    centroids[:, 1],
    s=200,
    c='red',
    label='Centroids'
)

plt.title('K-Means Clustering (Iris Dataset)')
plt.xlabel('Feature 1')
plt.ylabel('Feature 2')
plt.legend()
plt.show()

```



COMPARISON

```

from sklearn.metrics import confusion_matrix

print(confusion_matrix(df['Species'].astype('category').cat.codes, y_kmeans))

[[ 0 50  0]
 [39  0 11]
 [14  0 36]]

```

